



TRACE: Triplet Recovery via Arrowhead-Centric Extraction for Flowchart Understanding

Lenovo

Daozhu Dong¹, Kaiwen Shi¹, Dan Si², Xiaoyu Hao², Tong Liu², Wenjie Zhang², Gong Cheng¹

¹ State Key Laboratory of Novel Software Technology, Nanjing University, Nanjing, China

² Lenovo (Beijing) Co., Ltd., Beijing, China

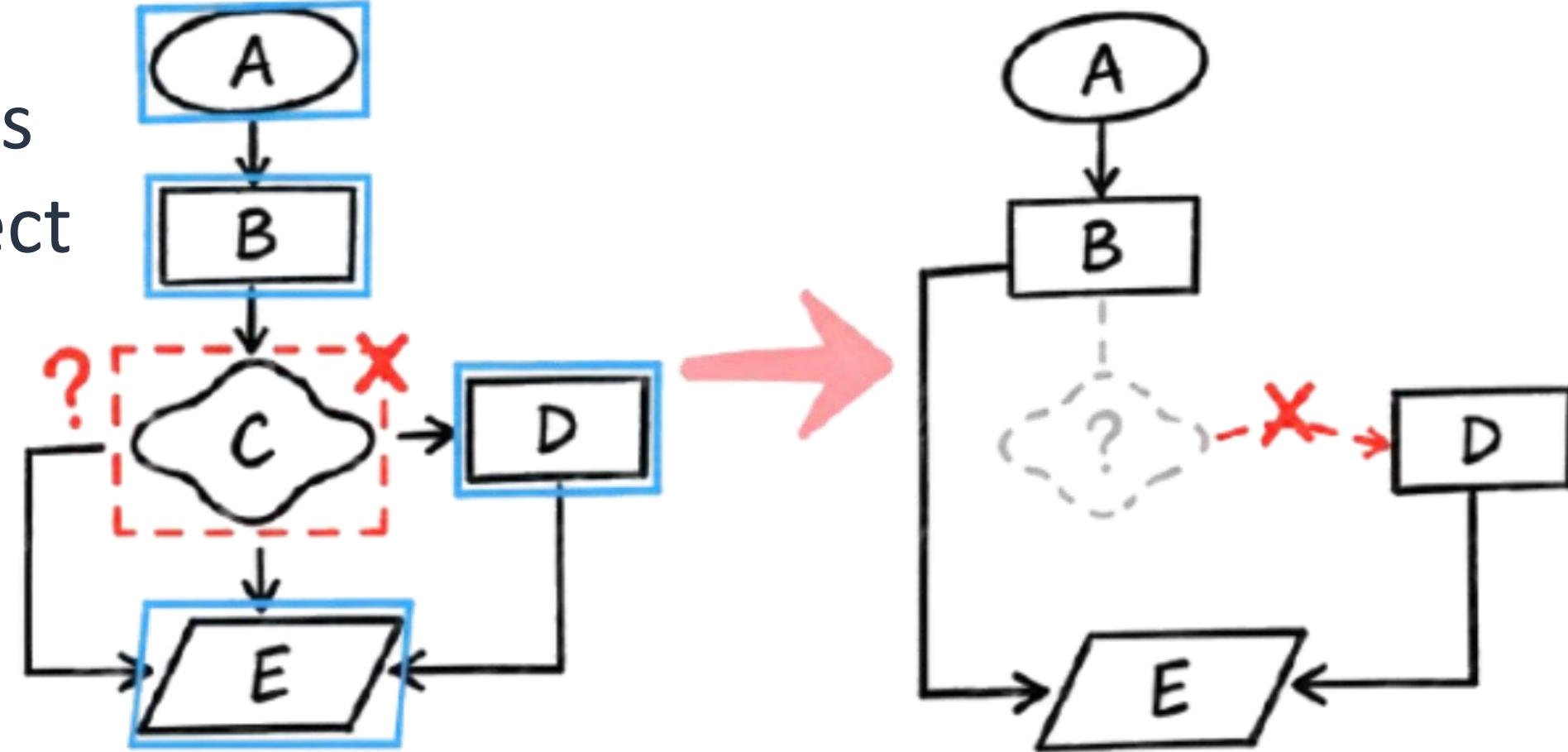
1

Motivation

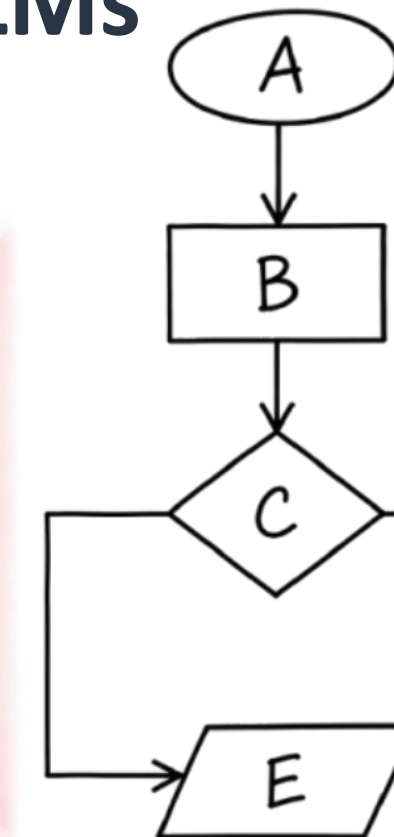
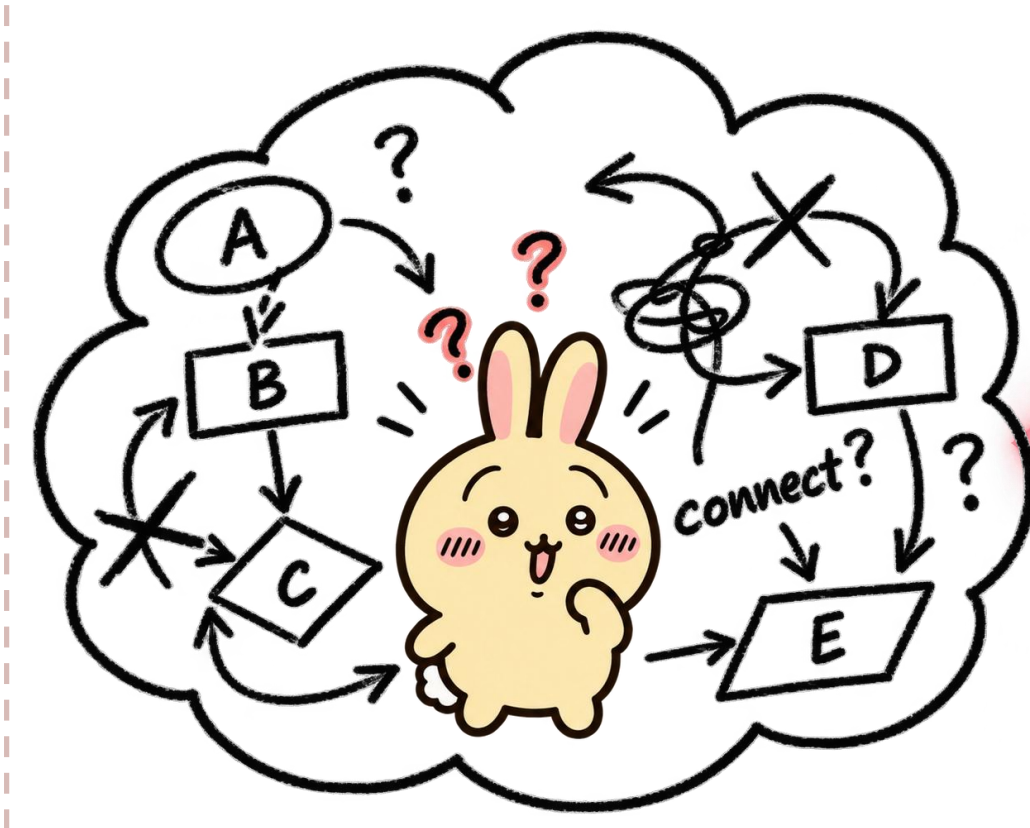
Why are flowchart edges difficult to recover?

Detection → Reconstruction

A missed shape leads to missing or incorrect edges.



Whole-image VLMs

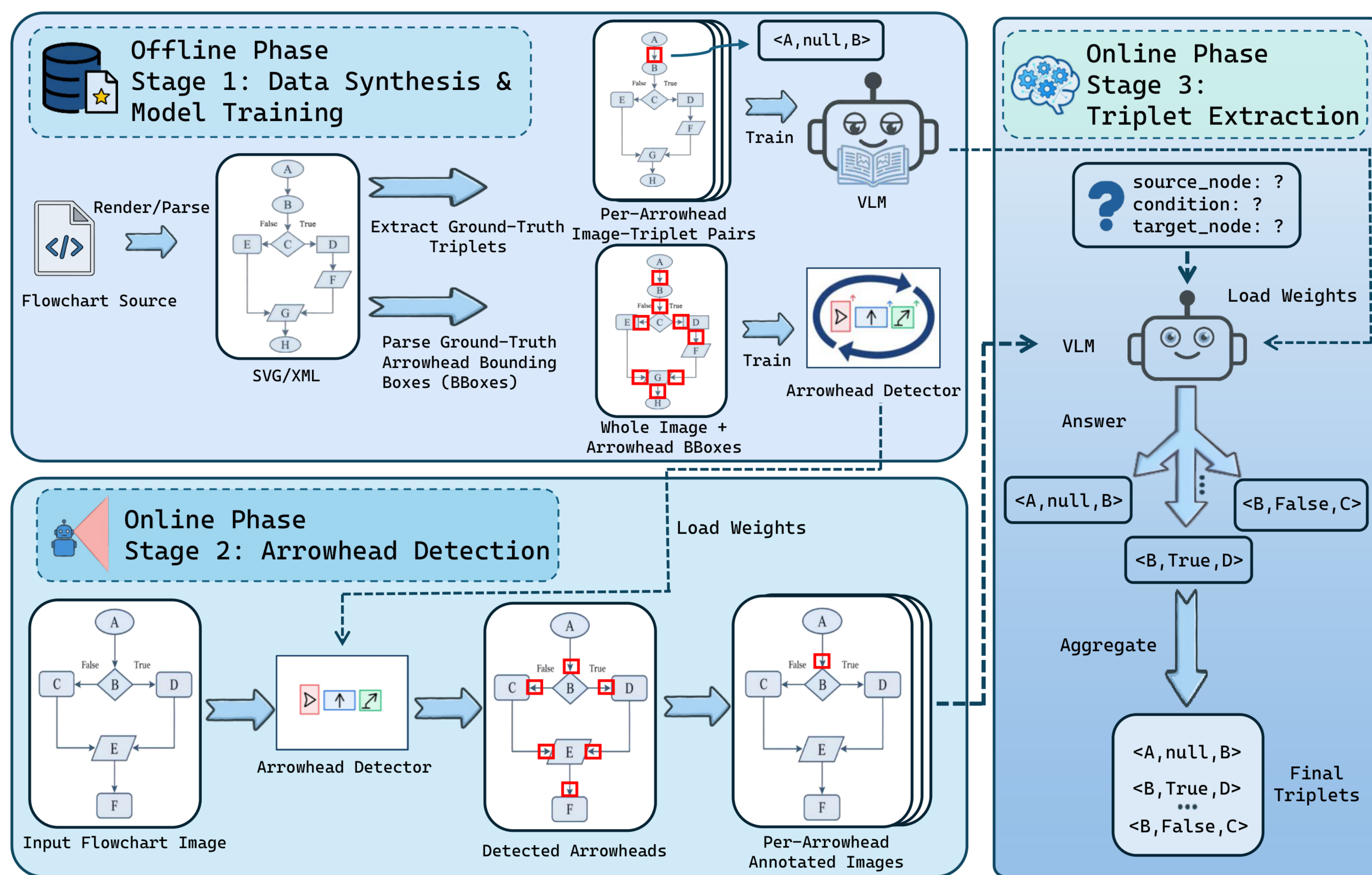


Dense flowcharts make individual arrows easy to miss or reverse.

2

Method

TRACE: One arrowhead per query, with the full image as context



1 Data synthesis & training

Render and parse native flowchart files (e.g., SVG/XML) to generate arrowhead boxes and per-arrowhead image-triplet pairs. Use them to train the detector and VLM.

2 Detect and highlight

Locate arrowheads and mark one per copy of the full image.

3 Trace and aggregate

Recover (source, condition, target). Aggregate outputs into a graph.

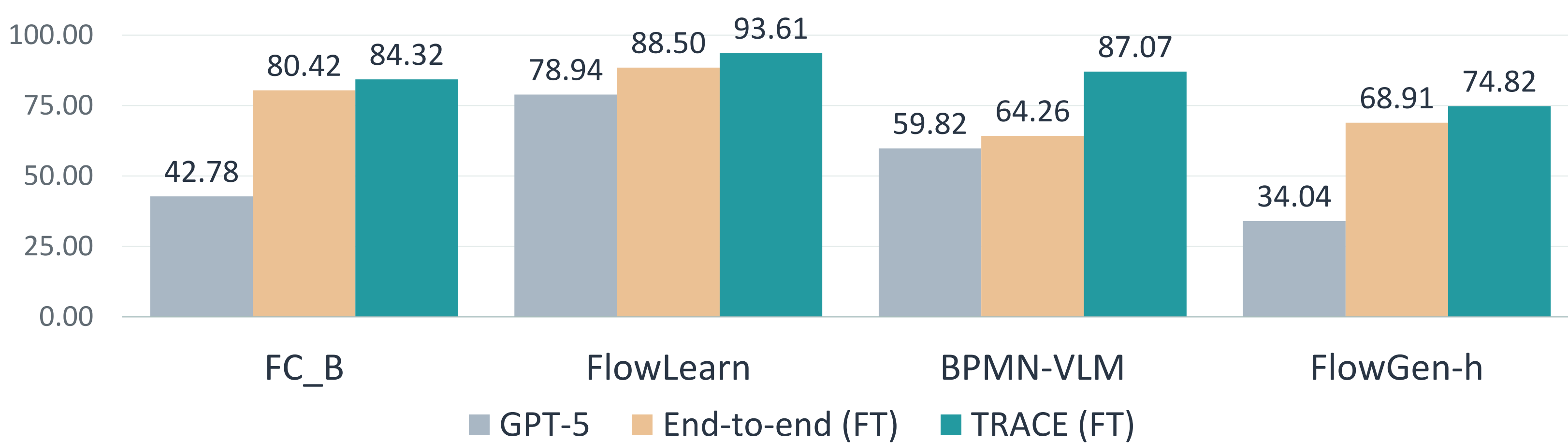
Every output points back to an arrow.

3

Results

Triplet extraction F1 (%)

FT: Qwen3-VL-4B. GPT-5: high reasoning, 3-shot.



+22.81 F1 on BPMN-VLM vs. End-to-end (FT)

Downstream QA accuracy (%)

Same Qwen3-VL-4B reasoner

TextFlow receives ground-truth Mermaid.

Method	FlowLearn	FlowVQA
Qwen3-VL-4B (ZS)	66.83	62.03
TextFlow + GT text	88.33	75.18
TRACE + tool-call	98.17	75.78
GT triplets (oracle)	100.00	78.46

+9.84 pp on FlowLearn vs. TextFlow

4

Analysis

How robust is TRACE across domains?

Higher F1 on 8/9 unseen benchmarks.

Train on 8 benchmarks, test on the held-out one. F1 (%).

Benchmark	E2E LOO	TRACE LOO	Gain
FlowVQA	87.20	89.31	+2.11
CBD	83.66	83.31	-0.35
FC_A	45.46	61.38	+15.92
FC_B	57.26	67.76	+10.50
FlowLearn	53.77	65.93	+12.16
BPMN-VLM	31.86	39.18	+7.32
FlowGen-e	95.87	96.97	+1.10
FlowGen-m	65.21	79.55	+14.34
FlowGen-h	57.93	69.42	+11.49

Qwen3-VL-4B. E2E = end-to-end. No OCR post-processing. Distinct formats remain challenging.

Is one query per arrowhead too slow?

Batch 3 arrowheads for 39% lower latency.

K = arrowheads per query. Macro averages over 9 benchmarks.

Extraction accuracy (%)

Method	F1	rF1
End-to-end	78.39	82.02
K = 1	83.30	86.52
K = 2	82.92	86.08
K = 3	82.81	85.84

Latency (seconds / image)

End-to-end	2.73
K = 1	4.00
K = 2	2.78
K = 3	2.44

K = 3: 1.64× faster than K = 1, with only 0.49 F1 loss.

FlowVQA latency: 11.41 s to 4.58 s (2.49× faster).

rF1 = relaxed F1. Qwen3-VL-4B on NVIDIA RTX 5880 Ada.

Key terms

- ZS** Zero-shot: no task-specific fine-tuning.
- FT** Fine-tuned on flowchart training data.
- LOO** Leave-one-out: train on 8 benchmarks, test on the 9th.

Correspondence: gcheng@nju.edu.cn

Code & data

github.com/nju-websoft/TRACE

Scan to explore the project

